INCH-POUND
MIL-S-28714B(ES)
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MILITARY SPECIFICATION

SWITCHES, ROTARY, SOLENOID OPERATED, (SWITCHBOARD-SELECTOR), GENERAL SPECIFICATION FOR

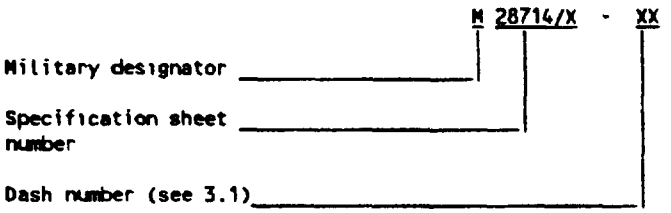
This specification is approved for use by the Defense Electronics Supply Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for solenoid operated, switchboard-selector, rotary switches. As used in this specification, the term "switches" is considered to mean the complete assembly.

1.2 Classification.

1.2.1 Military Part or Identifying Number (PIN). The term "Part or Identifying Number" (PIN) is equivalent to the term "part number" which was previously used in this specification. The military PIN consists of the letter "M", the basic number of the specification sheet, and an assigned dash number (see 3.1), as shown in the following example:



2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (6.2).

* FEDERAL SPECIFICATIONS

QQ-S-571 - Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DESC-ELDM, Defense Electronics Supply Center, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* MILITARY SPECIFICATIONS

MIL-M-14/6 - Molding Compounds, Diallyl Ortho-Phthalate, Thermosetting, Containing Mineral Fillers, Glass or Polymeric Fillers.

MIL-S-4040 - Solenoid, Electrical, General Specification for.

MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.

MIL-S-13949/4 - Sheet, Printed Wiring Board, Laminated, Base Material GF (Woven E-Glass Reinforcement, Majority Difunctional Epoxy Resin, Flame Resistant, Metal Clad or Unclad).

MIL-S-28714/1 - Switch, Rotary, Solenoid Operated, (Switchboard-Selector), Switch Assembly.

MIL-S-28714/2 - Switch, Rotary, Solenoid Operated, (Switchboard-Selector), Driver Unit.

MIL-S-28714/3 - Switch, Rotary, Solenoid Operated, (Switchboard-Selector), Switch Section Assembly.

MIL-S-28786 - Switches, Electrical and Fiber Optic, Packaging of.

MIL-G-45204 - Gold Plating, Electrodeposited.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services.

* MILITARY STANDARDS

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

MIL-STD-889 - Dissimilar Metals.

MIL-STD-1285 - Marking of Electrical and Electronic Parts.

MIL-STD-45662 - Calibration Systems Requirements.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specifications sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.5.

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* 3.3 Material. Material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the switches to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

* 3.3.1 Stator insulation. Insulation material shall be type GF, .062 inch thick, clad on each side with 2 oz/ft^2 copper, in accordance with MIL-S-13949/4.

3.3.2 Fungus resistance. All insulation materials shall be nonfungus nutrient in accordance with MIL-STD-454, requirement 4. In addition, certification for paper-base laminated materials shall be based upon parts that are in finished condition and ready for assembly into the final product.

3.3.3 Terminal boards. Terminal boards, when used, shall be of GDI-30 glass fiber reinforced compound material in accordance with MIL-M-14/6.

3.3.4 Plating. Plating on the printed wiring circuitry shall be gold plate, with a minimum of .00005 inch and a maximum of .0001 inch thick, and shall be in accordance with MIL-G-45204.

3.3.5 Metals. All metal parts shall be of corrosion-resistant material, or shall be suitably protected to resist corrosion. Solenoid high torque ball-race surfaces shall be of an industrial satin chrome type plating.

* 3.3.5.1 Dissimilar metals and compatible couples. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy) is not acceptable. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. The use of corrosion resistant steel in contact with cadmium plating is permitted. Dissimilar metals and compatible couples are defined in MIL-STD-889.

3.3.5.2 Solder. Solder, when used, shall be composition Sn60 in accordance with QQ-S-571.

* 3.3.5.3 Finish. Unless otherwise specified (see 3.1), all aluminum parts shall be anodized in accordance with MIL-A-8625, or chemically treated in accordance with MIL-C-5541.

* 3.3.6 Solenoid. The solenoid shall be rotary type designed in accordance with MIL-S-4040. The duty cycle, input voltage, and torque output shall be as specified (see 3.1).

3.4 Design and construction. Switches shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Threaded parts. Unless otherwise specified (see 3.1), all threaded parts shall be in accordance with FED-STD-H28. Wherever possible, the Unified Thread Series shall be used. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch which is used in the fine-thread series.

3.4.1.1 Locking of screw-thread assemblies. All screw-thread assemblies shall be rendered resistant to loosening under vibration. All threaded holes of aluminum shall have Heli-coil (or equal) inserts, except the threaded holes in the rails for connector locking means.

3.4.2 Contacts. Contacts shall be self-cleaning, and shall be of such type that no adjustments are required during the life of the switch.

3.4.2.1 Contact material. Contact material shall be of silver basis metal, and shall have rolled inlay consisting of silver, gold, and platinum, .0006 inch thickness minimum. Movable rotor blade contacts shall be of the same material as the stationary contacts.

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3.4.2.2 Contact type. The main switch section shall incorporate contact type design switches wherein the contact rotor blade passes between two fixed stator contact members. The switch movable contacts shall consist of a rotor blade which passes between the fixed members of the stator contact, and which does not contact the surface of the switch wafer at any point during its rotation. The rotor shall be captive to the stator.

3.4.2.3 Contact alignment. In all switch positions, a plane normal to the switch sections and through the longitudinal centerlines of the stationary contacts being engaged, shall intersect their movable contacts.

3.4.3 Operating shaft. The operating shaft shall be metal and shall be insulated from all terminals. The length, shape, and special shaft construction shall be as specified (see 3.1). Unless otherwise specified (see 3.1), the operating shaft shall be .250 +.001, -.003 inch in diameter, and the double D flat shall be .185 +.001, -.000.

3.4.4 Indexing. Solenoid drive mechanism shall have a positive detent mechanism. Glass or other fragile materials shall not be used in indexing or detent mechanisms.

3.5 Circuit configuration. When switches are tested as specified in 4.7.2, the switch-circuit configurations shall conform to the applicable diagrams specified. Switches shall make and break the required circuits in all positions and in all sections.

* 3.6 Contact resistance. When measured as specified in 4.7.3, the contact resistance shall not exceed the following values, unless otherwise specified (see 3.1):

a. Initial and after vibration and shock tests: 75 milliohms.

b. After moisture resistance and salt spray tests: 85 milliohms.

c After life (rotational) tests: 100 milliohms.

During the test, there shall be no evidence of arcing or instability of the contacts.

3.7 Thermal shock. When switches are tested as specified in 4.7.4, there shall be no mechanical or electrical damage, loosening of rivets, or other fastening devices.

3.8 Vibration. When switches are tested as specified in 4.7.5, mating contacts shall not open for a period of time longer than 10 microseconds before reclosing, unless otherwise specified (see 3.1). After the test, there shall be no change in shaft position, or evidence of broken, loose, deformed, or displaced parts.

3.9 Shock (specified pulse). When switches are tested as specified in 4.7.6, mating contacts shall not open for a period of time longer than 10 microseconds, unless otherwise specified (see 3.1). After the test, there shall be no change in shaft position, or evidence of broken, loose, deformed, or displaced parts.

3.10 Dielectric withstanding voltage. When tested as specified in 4.7.7, switches shall withstand the application of the specified voltages without arcing, flashover, breakdown of insulation, or damage.

3.11 Insulation resistance. When measured as specified in 4.7.8, the insulation resistance shall be not less than 100 megohms.

3.12 Moisture resistance. When switches are tested as specified in 4.7.9, the insulation resistance, immediately after conclusion of the test and while the switches are still in the humidity chamber, shall be greater than 10 megohms, unless otherwise specified (see 3.1). At the end of the drying period, the insulation resistance shall be as specified in 3.11. At the conclusion of the test, there shall be no evidence of corrosion, breaking, cracking, spalling, or loosening of terminals outside the confines of the terminal anchoring device, and mounting hardware shall be readily removable.

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3.13 Salt spray (corrosion). When switches are tested as specified in 4.7.10, there shall be no evidence of excessive corrosion. Excessive corrosion is defined as corrosion which interferes with the electrical or mechanical performance, or, in the case of plated metals, corrosion which has passed through the plating and attacked the base metal. After the test, mounting hardware shall be readily removable.

3.14 Life (rotational). When tested as specified in 4.7.11, switches shall be electrically and mechanically operative for 500,000 rotational steps during and after test. During the test, there shall be no evidence of intermittent operation. Movement of terminals within the confines of their terminal anchoring devices shall not constitute a failure. The specified contact sequential pattern shall be maintained throughout the test.

3.15 Coil current. When switches are tested as specified in 4.7.12, the coil current shall not exceed the value specified (see 3.1).

3.16 Marking. Switches shall be marked in accordance with MIL-STD-1285, with the following information:

a. Military PIN (see 1.2.1).

* b. Switch manufacturer's name, trademark, or source code.

c. Switch manufacturer's part number.

3.17 Workmanship. Switches shall be processed in such a manner as to be uniform in quality and shall be free from cracked or displaced parts, and other defects that will affect life, serviceability, or appearance. All molded or laminated parts shall be free of chips, blemishes, or flakes which are detrimental to the operating function of the switch. Gate marks shall not be considered a defect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. First article inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

* 4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the switches are in accordance with the applicable referenced specifications or requirements, prior to such fabrication.

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4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-202.

4.5 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.5.1 Sample size. The number of sample units comprising a sample of switches to be submitted for first article inspection shall be as specified in table I.

4.5.2 Inspection routine. Sample units shall be subjected to the first article inspection specified in table I, in the order shown. All sample units shall be subjected to the inspections of group I. The sample units shall then be divided as specified in table I for groups II to V, inclusive, and subjected to the inspections for their particular group. Any switch failing any examination or test in the sequence shall not be subjected to further inspection.

4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I (4 sample units)		
Visual and mechanical examination	3.1, 3.3 to 3.4.4, 3.16, and 3.17	4.7.1
Circuit configuration	3.5	4.7.2
Contact resistance	3.6	4.7.3
Group II (1 sample unit)		
Thermal shock	3.7	4.7.4
Vibration	3.8	4.7.5
Shock (specified pulse)	3.9	4.7.6
Dielectric withstanding voltage	3.10	4.7.7
Insulation resistance	3.11	4.7.8
Contact resistance	3.6	4.7.3
Group III (1 sample unit)		
Moisture resistance	3.12	4.7.9
Dielectric withstanding voltage	3.10	4.7.7
Contact resistance	3.6	4.7.3
Group IV (1 sample unit)		
Insulation resistance	3.11	4.7.8
Salt spray (corrosion)	3.13	4.7.10
Contact resistance	3.6	4.7.3
Group V (1 sample unit)		
Life (rotational)	3.14	4.7.11
Coil current	3.15	4.7.12
Dielectric withstanding voltage	3.10	4.7.7
Insulation resistance	3.11	4.7.8
Contact resistance	3.6	4.7.3

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4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

* 4.6.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all the switches of the same specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

* 4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II.

* 4.6.1.2.1 Sampling plan. Statistical sampling shall be in accordance with tables II and III. Samples selected for subgroup 1 testing may also be used for subgroup 2 tests. For acceptance of the lot, there shall be zero occurrences of defects.

* 4.6.1.2.1.1 Subgroup 1. A sample of parts shall be randomly selected in accordance with table III, subgroup 1 sampling plan.

* 4.6.1.2.1.2 Subgroup 2. A sample of parts shall be randomly selected in accordance with table III, subgroup 2 sampling plan.

4.6.1.2.2 Disposition of sample units. Sample units which have passed all the group A inspection shall be delivered on the contract or order, if the lot is accepted.

* 4.6.1.2.3 Rejected lots. If an inspection lot is rejected, the lot shall be 100 percent inspected for the defects noted. The contractor may correct all of the defects or remove all of the defective units from the lot. The lot shall then be sampled again in accordance with tables II and III. For acceptance, there shall be zero occurrences of defects. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

* TABLE II. Group A inspection

Inspection	Requirement paragraph	Method paragraph	Sampling plan
Subgroup 1			4.6.1.2.1.1 1/
Visual and mechanical examination		4.7.1	
Material	3.1 and 3.3 to 3.3.6 inclusive		
Design and construction	3.4 to 3.4.4 inclusive		
Marking	3.16		
Workmanship	3.17		
Dielectric withstanding voltage	3.10	4.7.7	
Contact resistance	3.6	4.7.3	
Subgroup 2			4.6.1.2.1.2
Circuit configuration	3.5	4.7.2	

1/ At the option of the contractor, in-process inspection may be used to meet the materials and design and construction requirements provided that they meet the acceptance criteria and all of the contractor's in-process control data on these tests are made available to the Government upon request.

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TABLE III. Zero defect sampling plan.

Lot size	Number of switches to be tested	
	Subgroup 1	Subgroup 2
1 - 8	3	All
9 - 15	3	13
16 - 25	3	13
26 - 50	5	13
51 - 90	6	13
91 - 150	7	13
151 - 280	10	20
281 - 500	11	29
501 - 1,200	15	34
1,201 - 3,200	18	42
3,201 - 10,000	22	50
10,001 - 35,000	29	60

4.6.2 Packaging inspection. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of MIL-S-28786.

4.7 Methods of inspection.

4.7.1 Visual and mechanical examination. Switches shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (See 3.1, 3.3 through 3.4.4, 3.16, and 3.17.)

4.7.2 Circuit configuration (see 3.5). Switches shall be tested with suitable test circuits to determine conformance of the actual open and closed switch-circuit conditions, including shorting and nonshorting functions, to the applicable diagrams (see 3.1).

4.7.3 Contact resistance (see 3.6). Switches shall be tested in accordance with method 307 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of connection: To two different pairs of mating contacts on each electrical sector. Terminal edge connectors shall be used to make contact with the terminals of the switch card. Where the switch has undergone a life (rotational) test (see 4.7.11), the contacts shall have been subjected to the specified electrical test loads Contact resistance values obtained include the resistance of the printed circuitry and the external connectors, as well as the resistance of the mating contacts.

b. Test current: 100 milliamperes.

c. Maximum open-circuit test voltage: 2 volts.

d. Number of activations prior to measurement: Four, for mating contacts wiped across each other.

e. Number of test activations: One for each mating contact under test.

f. Number of measurements per activation: One for each pair of mating contacts under test, except that a minimum of four shall be made on each switch when possible.

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4.7.4 Thermal shock (see 3.7). Switches shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

a. Test condition letter: A (air, -55°C to +85°C).

b. Measurements before and after cycling: Not applicable.

c. Examinations after test: Switches shall be examined for mechanical and electrical damage and loosening of rivets or other fastening devices.

4.7.5 Vibration (see 3.8). Switches shall be tested in accordance with method 201 of MIL-STD-202 (10 to 55 Hz). The following details and exceptions shall apply:

a. Mounting: Switches shall be rigidly mounted by their normal mounting means. The mounting fixture shall be free from resonances over the test frequency range.

b. Direction of motion: One of the directions of vibration shall be in a plane perpendicular to the longitudinal axis of the switch.

c. Electrical load conditions: Unless otherwise specified (see 3.1), the test circuit shall consist of one pair of mated contacts in closed position on each electrical sector (each pole group of contacts that performs a nonduplicating electrical function), series-connected, or individually monitored with each switch under test set at a different operating position.

d. Test and measurement during vibration: Switch-contact stability shall be continuously monitored in accordance with method 310 of MIL-STD-202. Connections to the terminal switch cards may be made by means of terminal edge connectors. In the event of indication of failure, the vibration cycle shall be continued long enough to monitor contacts, switch by switch, and section by section, to determine if a switch is defective. If no opening can be found in a switch by switch or section by section monitoring, the switch may be considered to be a good unit.

e. Tests and measurements after vibration: Not applicable.

f. Examinations after test: Switches shall be examined for change in shaft position, and evidence of broken, deformed, displaced, or loose parts.

4.7.6 Shock (specified pulse) (see 3.9). Switches shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting means: Switches shall be rigidly mounted by their normal mounting means.

b. Test condition letter: C (100 g, half-sine).

c. Electrical load conditions: As specified in 4.7.5c.

d. Measurement during shock: Switch contact stability shall be continuously monitored during shock by means of a test circuit in accordance with method 310 of MIL-STD-202, and shall consist of one pair of mated contacts in closed position on each electrical sector (each pole group of contacts that performs a nonduplicating electrical function), series-connected, or individually monitored with each switch under test set at a different operating position. In the event of indication of contact opening greater than that allowed, the test shall be modified by applying successive identical blows in the same plane to monitor contacts, switch by switch, or section by section, to determine if a switch is defective. Connections to the terminal switch cards may be made by means of terminal edge connectors.

e. Examinations after test: Switches shall be examined for change in shaft position, and evidence of broken, deformed, displaced, or loose parts.

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4.7.7 Dielectric withstanding voltage (see 3.10). Switches shall be tested in accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

a. Magnitude of test voltage: 1,000 volts.

b. Nature of potential: DC.

c. Duration of application of test voltage: Slowly increase voltage, hold for 5 seconds then return to 0.

d. Points of application of test voltage for each switch section: Position switch to the off position. Connect test connectors (see figures 1 and 2 and table IV) to the first switch deck to be tested, then connect high voltage power supply to groups 1 and 2 terminals of test connectors (see figures 1 and 2) and test per table IV.

e. Examinations after test: Switches shall be examined for evidence of arcing, flashover, breakdown of insulation, and damage.

TABLE IV. Dielectric withstanding voltage test points.

Test	Deck	Connector	Group 1 red	Group 2 blue
1 1/	1, 2, 3, and 4	A	B, D, F, J, L, N, P, R, T, V, X, Z, BB, and all numbered terminals	A, C, E, H, K, M, S, U, W, Y, AA
		C	B, D, F, J, L, N, P, R, T, V, X, Z, BB, and all numbered terminals	A, C, E, H, K, M, S, U, W, Y, AA
		B	A, B, C, D, E, H, J, K, M, N, P, R, S, 1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13, 14, 15	F, L
2 2/	5, 6, 7, 8, 9, and 10	A	B, D, F, J, L, N, P, R, T, V, X, Z, BB, 2, 4, 6, 8, 10, 12, 13, 14, 16, 18, 20, 22, 24	A, C, E, H, K, M, S, U, W, Y, AA, 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23
		C	B, D, F, J, L, N, P, R, T, V, X, Z, BB, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24	A, C, E, H, K, M, S, U, W, Y, AA, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23
		B	A, B, C, D, E, H, J, K, M, N, P, R, S, 1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13, 14, 15	F, L, 6, 10

1/ Test 1 - On A and C connector, blue jumper no connection.
- On B connector, clip input only to jumper on F and L

2/ Test 2 - On A and C connector, connect blue jumper to blue group.
- On B connector, clip input to both jumpers F, L and 6, 10.

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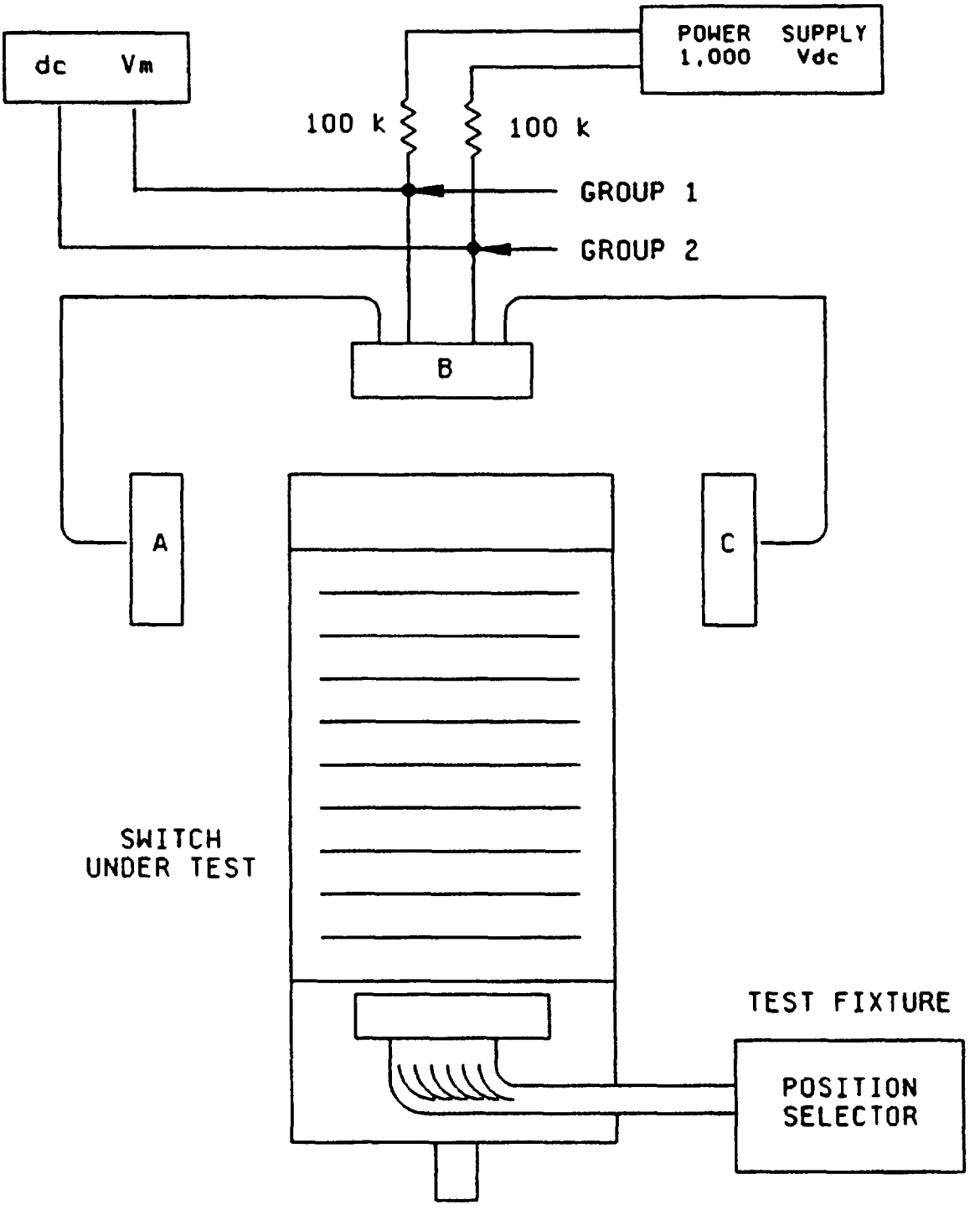


FIGURE 1. Dielectric withstanding voltage test hook-up.

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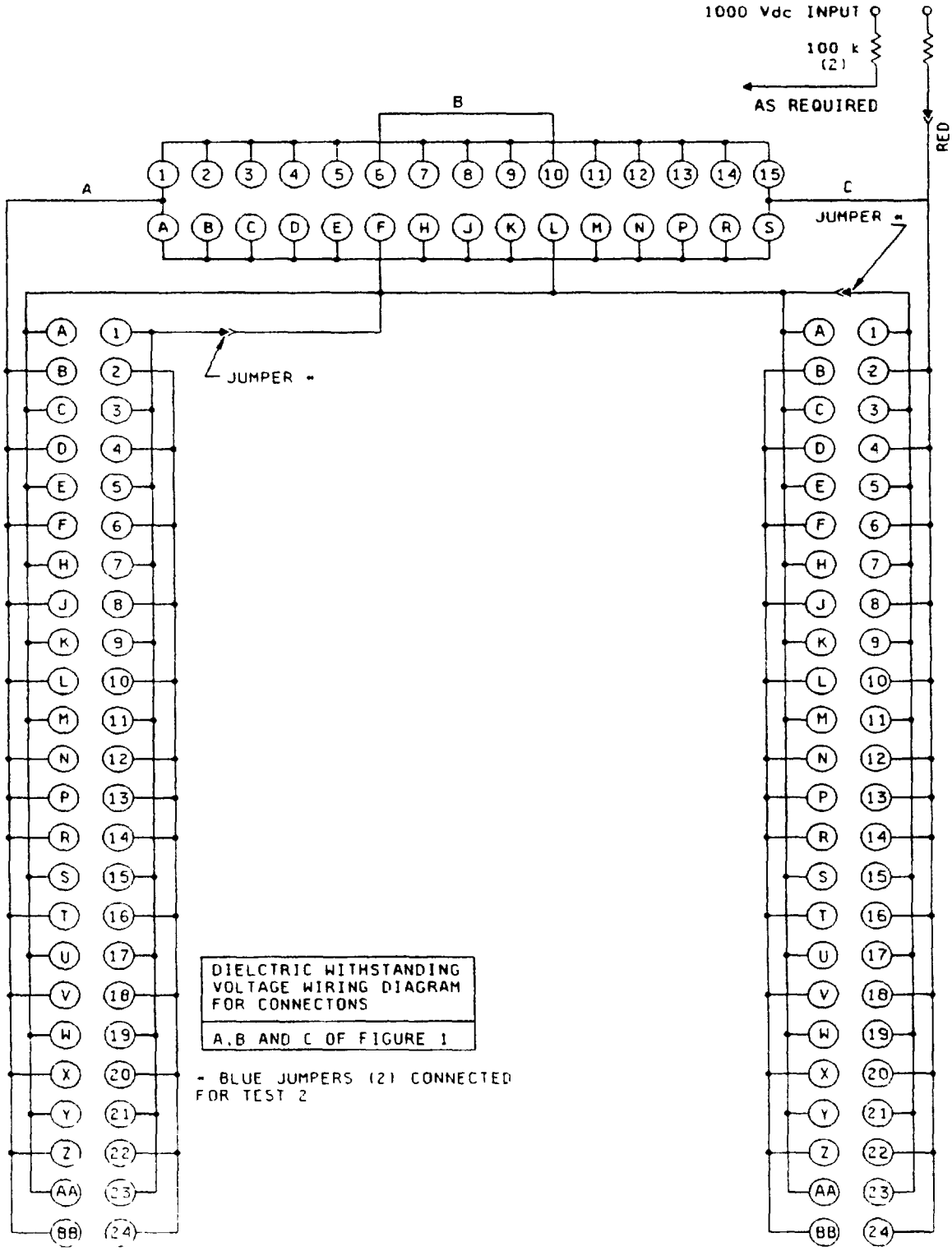


FIGURE 2. Wiring diagram

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4.7.8 Insulation resistance (see 3.11). Switches shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test condition letter: A (100 volts).

b. Points of measurement: Position switch to the off position. Install appropriate connector on switch deck 1, connector B. Connect one lead of megohm-meter to switch frame and the other lead as directed in table V. Repeat the measurement for each connector on a deck, and for all 10 decks.

TABLE V. Insulation resistance to ground.

Deck	Connector	Terminals
1, 2, 3, and 4	B	F and L
5, 6, 7, 8, 9, and 10	B	E, F, L, N, 4, 6, 10, and 12
1, 2, 3, and 4	A and C	A, C, E, H, K, M, S, V, W, Y, and AA
5, 6, 7, 8, 9, and 10	A and C	All terminals except N, P, 12, and 13

4.7.9 Moisture resistance (see 3.12). Switches shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Switches shall be mounted with the shaft in a horizontal position.

b. Initial measurement: Not applicable.

c. Polarization: During steps 1 to 6, inclusive, a polarization voltage of 100 volts dc shall be applied between the terminals as shown in figure 3.

d. Steps 7A and 7B: Not applicable.

e. Final measurements: Immediately after conclusion of the test and while the switches are still in the humidity chamber, insulation resistance shall be measured as specified in 4.7.8, except measurements shall only be made on those leads shown in figure 3. At the end of the drying period, insulation resistance shall again be measured as specified in 4.7.8.

f. Examination after test: Switches shall be examined for evidence of corrosion.

4.7.10 Salt spray (corrosion) (see 3.13). Switches shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply:

a. Test condition letter: B (48 hours).

b. Examination after test: Switches shall be examined for evidence of excessive corrosion.

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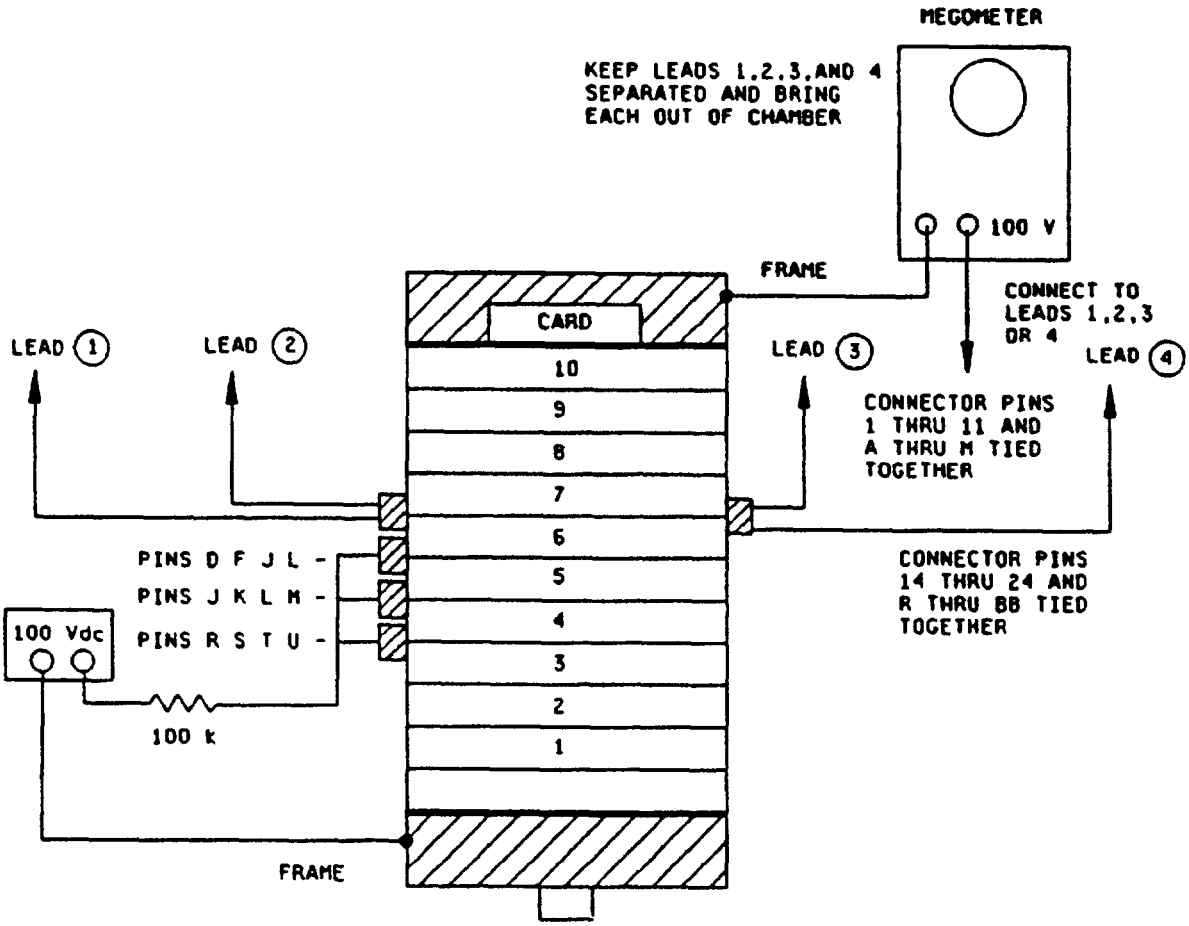


FIGURE 3. Moisture resistance test set-up.

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4.7.11 Life (rotational) (see 3.14). Switches shall be tested as follows:

a. Test potential and load: A minimum of 24 insulated sections shall be energized with the specified rated loads (see table VI), and the number of sections tested shall be as follows:

6 insulated sections: Inductive load.
6 insulated sections: Resistive load with lowest voltage.
12 insulated sections: Resistive load with highest voltage.

The minimum input voltage to the solenoid coil shall be as specified (see 3.1). The specified loads for the sections shall be wired to the contacts which are subject to making and breaking the circuit during rotation of the switch shaft:

Nonshorting type section: All contacts.

Shorting type section: Alternate contacts only, or those contacts which are subject to making and breaking during rotation of the switch shaft.

Suitable means shall be provided to indicate when contacts have failed to make and break in accordance with the specified circuit configuration.

b. Rotation and rotational rate: 48 steps per minute. At the option of the contractor, rotation may be interrupted, periodically, after each 120,000 or 166,668 steps of rotational operation. Rotation shall be in one direction only.

* c. Test condition: One sample shall be tested for 500,000 rotational steps, divided for the following three temperature conditions:

166,667 steps: at 0°C.

166,666 steps: at 25°C.

166,667 steps: at 50°C.

d. No supplementary lubricant shall be used.

e. Tests during rotations: One terminal of a suitable resistor shall be connected to the back contacts of the rear switch card, the other terminal of the resistor shall be connected to a neon lamp, and then the circuitry completed. At the beginning and end of each temperature test, the switches shall be monitored for 240 steps of operation for extinguishing of the neon lamp. Rotation of the switch may be slowed to allow observation of extinguishing of the neon lamp.

f. Examinations after test: Switches shall be examined for broken, deformed, displaced, or loose parts.

4.7.12 Coil current (see 3.15). With the unit stabilized at 20°C, +5°C, -0°C, unless otherwise specified (see 3.1), the maximum specified operating voltage shall be applied to the coil terminals, and within 10 seconds after application of the coil voltage, the coil current shall be measured.

TABLE VI. Circuit values for life (rotational) test.

Environmental condition	Inductive load (2.8 henries)		Resistive load	
	Milliamperes	Volts, dc	Milliamperes	Volts
At atmospheric pressure	20	30	200 50	30 dc 150 ac

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-28786.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The switches covered by this specification are designed for Radar Signal Distribution Switchboards SB/1505/SP, SB/1109/SP and SB/1108/SP.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. Title, number, and date of the applicable specification sheet.
d. The PIN for the item.
e. When a first article is required for inspection and approval (see 3.2 and 6.3).

* 6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.5.2. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. For the purpose of this specification, the following definitions apply:

6.4.1 Switch section. A switch section is that part of a switch consisting of a rotor and a stator assembly as a unit with the insulating material.

6.4.2 Electrical section. An electrical section is an assembly of elements performing the same electrical function as a switch section. When two units are mounted on opposite sides of a single physical section and electrically insulated from each other, either side may be considered an insulated section.

6.4.3 Rotational step. A rotational step consists of one energization of the solenoid, and one deenergization of the solenoid.

6.4.4 Switch card. A switch card is that part of a switch complete in itself, except for means of operating it, and consisting of four rotors and eight insulated stator sides assembled as a unit with insulating material.

* 6.5 Subject term (key word) listing.

Coil operated
Electro-mechanical actuation
SB/1108/SP
SB/1109/SP
SB/1505/SP
Signal distribution
Stepping
Switch assembly

* 6.6 Part or Identifying Number (PIN). See 1.2.1

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6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-28714B(ES)	2. DOCUMENT DATE (YYMMDD) 94-01-21

3. DOCUMENT TITLE
Switches, Rotary, Solenoid Operated, (Switchboard Selector)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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8. PREPARING ACTIVITY

a. NAME Defense Electronics Supply Center	b. TELEPHONE (Include Area Code) (1) Commercial (513) 296-6179 (2) DSN 986-6179
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